```c
1. // Write a C program to find the maximum and minimum elements in an array.
```

```c
#include <stdio.h>
int main()
{
    int n, min, max;
    printf("enter the size of array:");
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", arr[i]);

    min=max = arr[0];
    for (int i = 1; i < n; i++)
    {
        if (arr[i] < min) min = arr[i];
        if (arr[i] > max) max = arr[i];
    }
    printf("minimum:%d\n", min);
    printf("maximum: %d\n", max);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q1.c -o q1 } ; if ($?) { .\q1 }
enter the size of array:2

s
minimum:16
maximum: 4199623
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
2. // Implement a C program to reverse the elements of an array.
```

```c
#include <stdio.h>
int main(void){
    int n;
    printf("Enter the size of array: ");
    scanf("%d", &n);

    int arr[n];
```

```c
    printf("Enter %d elements: \n", n);
    for(int i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("\nReversed array: \n");
    for(int i = n - 1; i >= 0; i--){
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q2.c -o q2 } ; if ($?) { .\q2 }
Enter the size of array: 4
Enter 4 elements:
bbbb

Reversed array:
32 0 4214910 1961967888
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

3.

```c
// Write a C program to sort an array of integers in ascending order using a sorting algorithm of your choice (e.g., bubble sort, selection sort, insertion sort).

#include <stdio.h>

int main() {
    int n;
    printf("Enter the size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d elements: \n", n);
    for(int i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }
    // Bubble sort
```

```c
    for(int i = 0; i < n-1; i++){
        for(int j = 0; j < n-i-1; j++){
            if(arr[j] > arr[j+1]){
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
    printf("The elements of array in ascending order are: ");
    for(int i = 0; i < n; i++){
        printf("%d ", arr[i]);
    }
    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q3.c -o q3 } ; if ($?) { .\q3 }
Enter the size of array: 2
Enter 2 elements:
6
bb
The elements of array in ascending order are: 6 4199623
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
4.// Write a C program to merge two sorted arrays into a single sorted array.

#include <stdio.h>

int main() {
    int n1, n2;

    printf("Enter size of first and second array: ");
    scanf("%d %d", &n1, &n2);

    int arr1[n1], arr2[n2];

    printf("Enter value for first array: ");
    for(int i=0; i<n1; i++) scanf("%d", &arr1[i]);

    printf("Enter value for second array: ");
    for(int j=0; j<n2; j++) scanf("%d", &arr2[j]);

    int merged_arr[n1+n2];
```

```c
    int x=0, y=0, z=0;

    while (x<n1 || y<n2) {
        if (arr1[x] < arr2[y]) merged_arr[z++] = arr1[x++];
        else merged_arr[z++] = arr2[y++];
    }

    while (x<n1) merged_arr[z++] = arr1[x++];
    while (y<n2) merged_arr[z++] = arr2[y++];

    printf("Merged array: ");
    for(int k=0; k<n1+n2; k++) printf("%d ", merged_arr[k]);
    printf("\n");

    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q5.c -o q5 } ; if ($?) { .\q5 }
Enter size of first and second array: 5
8
Enter value for first array: basant
Enter value for second array: Merged array: 4199136 4214923 0 32 6422192 2 6422116 1419454544 4199631 4214923 6422220 6422216 6422268
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
5.// Write a C program to find the length of a string without using the built-in string functions.

#include <stdio.h>
int main() {
    char abc[100];
    int count = 0;

    printf("enter the string:");
    scanf("%s", &abc);

    for (int i=0; i<=100; i++) {
        if (abc[i] == '\0') {
            break;
        }
        count++;
    }
```

```
    printf("length of string is %d.\n", count);
    return 0;}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q6.c -o q6 } ; if ($?) { .\q6 }
enter the string:4
length of string is 1.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q6.c -o q6 } ; if ($?) { .\q6 }
enter the string:
```

```
6.// Write a function named findFactorial that takes an integer as input and returns its factorial.



#include<stdio.h>

int findfactorial(int num) {
   int i=1, fact = 1;
    while(i<=num) {

     fact *=i;
     i++;
    }
return fact;
}



int main() {

    int num;
    printf("enter the number:");
    scanf("%d", &num);
```

```c
    int factorial = findfactorial (num);

    printf("the factorial is %d.\n", factorial);

return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q13.c -o q13 } ; if ($?) { .\q13 }
enter the number:4
the factorial is 24.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
7.// Write a function named calculatePower that takes two integers, base and exponent, as input and returns the result of raising the base to the exponent.


#include <stdio.h>

int calculatePower(int base, int exponent) {
    int result = 1;
    for (int i=0;i<exponent; i++) {
        result *= base;
    }
    return result;
}

int main() {
    int b, e;
    printf("Enter base and exponent: ");
    scanf("%d %d", &b, &e);

    int power = calculatePower(b, e);
    printf("Result is: %d.\n", power);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q17.c -o q17 } ; if ($?) { .\q17 }
Enter base and exponent: 3 4
Result is: 81.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/8/ Implement a C program to find the second largest element in an array.

#include <stdio.h>

int main()
{
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int largest = arr[0];
    int secondLargest = -2147483648;

    for (int i = 1; i < n; i++)
    {
        if (arr[i] > largest)
        {
            secondLargest = largest;
            largest = arr[i];
        }
        else if (arr[i] > secondLargest && arr[i] != largest)
        {
            secondLargest = arr[i];
        }
    }

    if (secondLargest == -2147483648)
    {
```

```c
        printf("No second largest element exists.\n");
    }
    else
    {
        printf("Second largest element is: %d\n", secondLargest);
    }

    return 0;
}
```

```
sktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q4.c -o q4 } ; if ($?)
 { .\q4 }
Enter number of elements: 3
Enter elements:
abc
Second largest element is: 4199623
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/9/ Implement a C program to reverse a string.



#include <stdio.h>
#include <string.h>

int main() {
    char str[100];

    printf("Enter a string: ");
    scanf("%s", str);

    int length = strlen(str);


    for(int i = 0; i < length / 2; i++) {
        char temp = str[i];
        str[i] = str[length - i - 1];
        str[length - i - 1] = temp;
    }

    printf("Reversed string: %s\n", str);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q7.c -o q7 } ; if ($?) { .\q7 }
Enter a string: 4
Reversed string: 4
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/10/ Write a C program to check if a given string is a palindrome.



#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int isPalindrome = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    int length = strlen(str);

    for(int i = 0; i < length / 2; i++) {
        if(str[i] != str[length - i - 1]) {
            isPalindrome = 0;
            break;
        }
    }

    if(isPalindrome)
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\U
signment-3-basantchand\src\" ; if ($?) { gcc q8.c -o q8 } ; if ($?) { .\
Enter a string: 7
The string is a palindrome.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
11// Implement a C program to count the occurrence of a specific character in a string.


#include <stdio.h>
#include <string.h>

int main() {
    char str[100], ch;
    int count = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    printf("Enter the character to count: ");
    scanf(" %c", &ch);

    int length = strlen(str);

    for(int i = 0; i < length; i++) {
        if(str[i] == ch) {
            count++;
        }
    }

    printf("Character '%c' occurs %d times.\n", ch, count);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src> cd "c:\Users\ASUS\Desktop\Cprogrammin
g\assignment-3-basantchand\src\" ; if ($?) { gcc q9.c -o q9 } ; if ($?) { .\q9 }
Enter a string: 5
Enter the character to count: g
Character 'g' occurs 0 times.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/12/ Write a C program to concatenate two strings without using the built-in
string functions.

#include <stdio.h>

int main()
{
    char str1[100], str2[100];
```

```
    printf("Enter first string: ");
    scanf("%s", str1);

    printf("Enter second string: ");
    scanf("%s", str2);

    int i = 0, j = 0;

    while (str1[i] != '\0')
    {
        i++;
    }

    while (str2[j] != '\0')
    {
        str1[i] = str2[j];
        i++;
        j++;
    }

    str1[i] = '\0';

    printf("Concatenated string: %s\n", str1);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q10.c -o q10 } ; if ($?) { .\q10 }
Enter first string: 2
Enter second string: 4
Concatenated string: 24
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/13/ Write a function named calculateAverage that takes an array of integers as input and returns the average of the numbers.

#include <stdio.h>

float calculateAverage(int arr[], int size)
{
    int sum = 0;

    for (int i = 0; i < size; i++)
    {
        sum += arr[i];
```

```c
    }

    return (float)sum / size;
}

int main()
{
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    float avg = calculateAverage(arr, n);

    printf("Average = %.2f\n", avg);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\as
signment-3-basantchand\src\" ; if ($?) { gcc q11.c -o q11 } ; if ($?) { .\q11 }
Enter number of elements: 123
Enter elements:
abc
Average = 4793907.50
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/14/  Write a function named isPalindrome that takes a string as input and
returns 1 if it is a palindrome (reads the same forwards and backwards), and 0
otherwise.

#include <stdio.h>
#include <string.h>

int isPalindrome(char str[])
{
    int length = strlen(str);
```

```c
    for (int i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - i - 1])
        {
            return 0;
        }
    }
    return 1;
}

int main()
{
    char str[100];

    printf("Enter a string: ");
    scanf("%s", str);

    if (isPalindrome(str))
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q12.c -o q12 } ; if ($?) { .\q12 }
Enter a string: 4
The string is a palindrome.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/15/ Write a function named convertTemperature that takes a temperature value in Celsius and converts it to Fahrenheit. The function should return the converted temperature.



#include <stdio.h>

float convertTemperature(float celsius) {
    return (celsius * 9 / 5) + 32;
}
```

```
int main() {
    float celsius;

    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);

    float fahrenheit = convertTemperature(celsius);

    printf("Temperature in Fahrenheit: %.2f\n", fahrenheit);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q14.c -o q14 } ; if ($?) { .\q14 }
Enter temperature in Celsius: 3
Temperature in Fahrenheit: 37.40
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/16/ Write a function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.



#include <stdio.h>

int countOccurrences(char str[], char ch) {
    int count = 0;
    int i = 0;

    while(str[i] != '\0') {
        if(str[i] == ch) {
            count++;
        }
        i++;
    }

    return count;
}

int main() {
    char str[100], ch;

    printf("Enter a string: ");
```

```c
    scanf("%s", str);

    printf("Enter a character to count: ");
    scanf(" %c", &ch);

    int result = countOccurrences(str, ch);

    printf("Character '%c' occurs %d times.\n", ch, result);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q15.c -o q15 } ; if ($?) { .\q15 }
Enter a string: 7
Enter a character to count: 8
Character '8' occurs 0 times.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/17/ Write a function named reverseArray that takes an array of integers as input and reverses the order of the elements in the array.


#include <stdio.h>

void reverseArray(int arr[], int size) {
    int start = 0;
    int end = size - 1;

    while(start < end) {
        int temp = arr[start];
        arr[start] = arr[end];
        arr[end] = temp;

        start++;
        end--;
    }
}

int main() {
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];
```

```c
    printf("Enter elements:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    reverseArray(arr, n);

    printf("Reversed array:\n");
    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprog
signment-3-basantchand\src\" ; if ($?) { gcc q16.c -o q16 } ; if ($?) { .\q16 }
Enter number of elements: 7
Enter elements:
jjj
Reversed array:
4214911 4199740 16 32 0 4214911 1959674128
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/18/ Write a function named findPrimeNumbers that takes an integer n as input and
prints all prime numbers from 1 to n.

#include <stdio.h>
#include <stdbool.h>

bool isPrime(int num)
{
    if (num <= 1)
        return false;
    for (int i = 2; i * i <= num; i++)
    {
        if (num % i == 0)
            return false;
    }
    return true;
}

void findPrimeNumbers(int n)
{
    printf("Prime numbers from 1 to %d are:\n", n);
```

```c
    for (int i = 2; i <= n; i++)
    {
        if (isPrime(i))
        {
            printf("%d ", i);
        }
    }
    printf("\n");
}

int main()
{
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    findPrimeNumbers(n);
    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\
signment-3-basantchand\src\" ; if ($?) { gcc q18.c -o q18 } ; if ($?) { .\q18 }
Enter a number: 6
Prime numbers from 1 to 6 are:
2 3 5
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
// 19  Write a function named calculateFactorialSeries that takes an integer n as
input and prints the factorial series up to n.



#include <stdio.h>

unsigned long long factorial(int num) {
    unsigned long long fact = 1;
    for (int i = 1; i <= num; i++) {
        fact *= i;
    }
    return fact;
}

void calculateFactorialSeries(int n) {
    printf("Factorial series up to %d:\n", n);
    for (int i = 1; i <= n; i++) {
```

```c
        printf("%d! = %llu\n", i, factorial(i));
    }
}

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    calculateFactorialSeries(n);
    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogrammi
signment-3-basantchand\src\" ; if ($?) { gcc q19.c -o q19 } ; if ($?) { .\q19 }
Enter a number: 4
Factorial series up to 4:
1! = 1
2! = 2
3! = 6
4! = 24
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/20/ Write a function named calculateGCD that takes two integers as input and
returns their greatest common divisor (GCD).



#include <stdio.h>

int calculateGCD(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int main() {
    int num1, num2;
    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    int gcd = calculateGCD(num1, num2);
    printf("GCD of %d and %d is %d\n", num1, num2, gcd);
```

```
    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q20.c -o q20 } ; if ($?) { .\q20 }
Enter two integers: 2 3
GCD of 2 and 3 is 1
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/21/ Write a recursive function named calculateFactorial that takes an integer n as input and returns its factorial.



#include <stdio.h>

unsigned long long calculateFactorial(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * calculateFactorial(n - 1);
}


int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);

    unsigned long long fact = calculateFactorial(n);
    printf("Factorial of %d is %llu\n", n, fact);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q21.c -o q21 } ; if ($?) { .\q21 }
Enter a number: 7
Factorial of 7 is 5040
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/22/ Write a recursive function named calculateFibonacci that takes an integer n as input and returns the nth Fibonacci number. The Fibonacci sequence starts with 0 and 1, and each subsequent number is the sum of the two preceding numbers.



#include <stdio.h>

int calculateFibonacci(int n) {
    if (n == 0) {
        return 0;
    } else if (n == 1) {
        return 1;
    } else {
        return calculateFibonacci(n - 1) + calculateFibonacci(n - 2);
    }
}

int main() {
    int n;
    printf("Enter a positive integer: ");
    scanf("%d", &n);

    int fib = calculateFibonacci(n);
    printf("Fibonacci number F(%d) = %d\n", n, fib);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q22.c -o q22 } ; if ($?) { .\q22 }
Enter a positive integer: 2
Fibonacci number F(2) = 1
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/23/ Write a recursive function named calculateGCD that takes two integers a and b as input and returns their greatest common divisor (GCD).


#include <stdio.h>

int calculateGCD(int a, int b) {
```

```c
    if (b == 0) {
        return a;
    }
    return calculateGCD(b, a % b);
}

int main() {
    int num1, num2;
    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    int gcd = calculateGCD(num1, num2);
    printf("GCD of %d and %d is %d\n", num1, num2, gcd);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogrammi
signment-3-basantchand\src\" ; if ($?) { gcc q23.c -o q23 } ; if ($?) { .\q23 }
Enter two integers: 42
88
GCD of 42 and 88 is 2
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
//24 Write a recursive function named calculatePower that takes two integers base
and exponent as input and returns the result of raising the base to the exponent.



#include <stdio.h>

long long calculatePower(int base, int exponent) {
    if (exponent == 0) {
        return 1;
    } else if (exponent < 0) {
        return 0;
    }
    return base * calculatePower(base, exponent - 1);
}

int main() {
    int base, exponent;
    printf("Enter base and exponent: ");
    scanf("%d %d", &base, &exponent);
```

```c
    long long result = calculatePower(base, exponent);
    printf("%d^%d = %lld\n", base, exponent, result);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q24.c -o q24 } ; if ($?) { .\q24 }
Enter base and exponent: 2 4
2^4 = 16
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/25/ Write a recursive function named calculateSumOfDigits that takes an integer n as input and returns the sum of its digits.


#include <stdio.h>

int calculateSumOfDigits(int n) {
    if (n == 0) {
        return 0;
    }
    return (n % 10) + calculateSumOfDigits(n / 10);
}


int main() {
    int n;
    printf("Enter an integer: ");
    scanf("%d", &n);

    if (n < 0) {
        n = -n;
    }

    int sum = calculateSumOfDigits(n);
    printf("Sum of digits = %d\n", sum);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q25.c -o q25 } ; if ($?) { .\q25 }
Enter an integer: 4
Sum of digits = 4
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/26/ Write a recursive function named reverseString that takes a string as input and returns the reversed string.


#include <stdio.h>
#include <string.h>

void reverseString(char str[], int start, int end) {
    if (start >= end) {
        return;
    }

    char temp = str[start];
    str[start] = str[end];
    str[end] = temp;

    reverseString(str, start + 1, end - 1);
}

int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    str[strcspn(str, "\n")] = '\0';

    reverseString(str, 0, strlen(str) - 1);

    printf("Reversed string: %s\n", str);
    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q26.c -o q26 } ; if ($?) { .\q26 }
Enter a string: 4
Reversed string: 4
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/27/ Write a recursive function named printTriangle that takes an integer n as
input and prints a triangle of asterisks (*) with n rows.



#include <stdio.h>

void printStars(int count) {
    if (count == 0) {
        return;
    }
    printf("*");
    printStars(count - 1);
}

void printTriangle(int n) {
    if (n == 0) {
        return;
    }

    printTriangle(n - 1);

    printStars(n);
    printf("\n");
}


int main() {
    int n;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    printTriangle(n);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q27.c -o q27 } ; if ($?) { .\q27 }
Enter number of rows: 2
*
**
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
28// Write a recursive function named calculateBinary that takes an integer n as input and returns its binary representation as a string.


#include <stdio.h>
#include <string.h>

void calculateBinary(int n, char str[], int *index) {
    if (n == 0) {
        return;
    }

    calculateBinary(n / 2, str, index);

    str[*index] = (n % 2) + '0';
    (*index)++;
}

int main() {
    int n;
    printf("Enter an integer: ");
    scanf("%d", &n);

    if (n == 0) {
        printf("Binary: 0\n");
        return 0;
    }

    char binary[64];
    int index = 0;

    calculateBinary(n, binary, &index);
    binary[index] = '\0';
```

```c
    printf("Binary representation: %s\n", binary);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q28.c -o q28 } ; if ($?) { .\q28 }
Enter an integer: 44
Binary representation: 101100
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```c
/29/ Write a recursive function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (reads the same forwards and backwards), and 0 otherwise.


#include <stdio.h>
#include <string.h>

int isPalindrome(char str[], int start, int end) {
    if (start >= end) {
        return 1;
    }
    if (str[start] != str[end]) {
        return 0;
    }
    return isPalindrome(str, start + 1, end - 1);
}

int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    str[strcspn(str, "\n")] = '\0';

    int length = strlen(str);
    if (isPalindrome(str, 0, length - 1)) {
        printf("The string is a palindrome.\n");
    } else {
        printf("The string is NOT a palindrome.\n");
    }
```

```
    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                Code - src
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src\" ; if ($?) { gcc q29.c -o q29 } ; if ($?) { .\q29 }
Enter a string: 4
The string is a palindrome.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```

```
/30/  Write a recursive function named countOccurrences that takes a string and a character as input and returns the number of times the character appears in the string.


#include <stdio.h>
#include <string.h>

int countOccurrences(char str[], char ch, int index) {
    if (str[index] == '\0') {
        return 0;
    }
    int count = (str[index] == ch) ? 1 : 0;
    return count + countOccurrences(str, ch, index + 1);
}

int main() {
    char str[100];
    char ch;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    printf("Enter a character to count: ");
    scanf(" %c", &ch);

    int occurrences = countOccurrences(str, ch, 0);
    printf("The character '%c' appears %d times.\n", ch, occurrences);

    return 0;
}
```

```
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand> cd "c:\Users\ASUS\Desktop\Cprogramming\as
signment-3-basantchand\src\" ; if ($?) { gcc q30.c -o q30 } ; if ($?) { .\q30 }
Enter a string: 4
Enter a character to count: 2
The character '2' appears 0 times.
PS C:\Users\ASUS\Desktop\Cprogramming\assignment-3-basantchand\src>
```